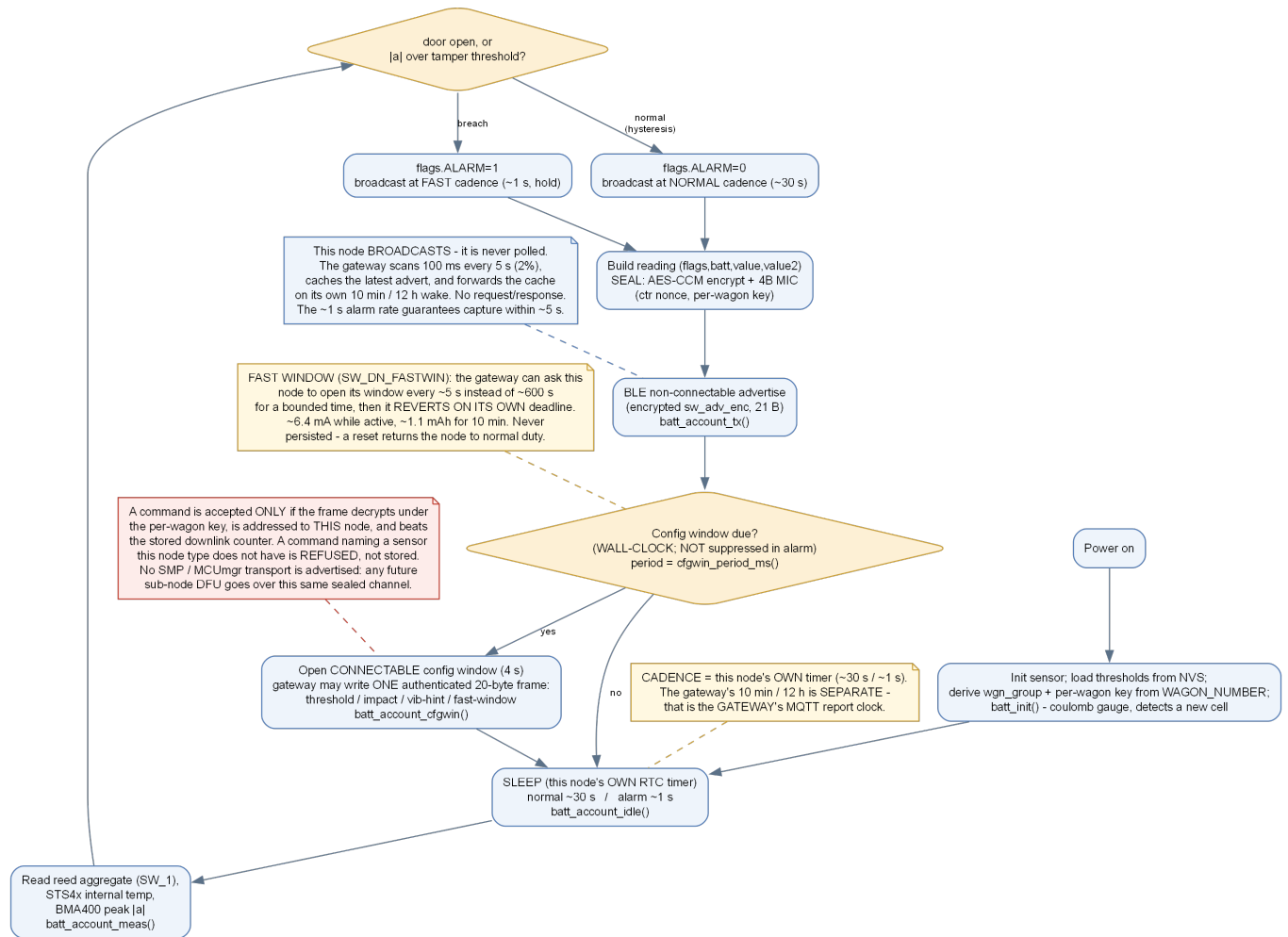


Sub-node — Door / Hatch Monitoring

4 per wagon · door state + tamper/impact + internal temperature



1. Role

A battery-powered, **connectionless BLE broadcaster**. It measures its sensor, **encrypts** the reading into a BLE advertisement (AES-CCM, with the wagon-group id for isolation), broadcasts a short burst, and sleeps. On a threshold breach it switches to fast "alarm" advertising so the gateway's receive window catches it within ~5 s. It forms a connection only during a periodic **config window**, and only to receive thresholds.

Timing — no polling: this node advertises on its **own** timer — about every **30 s** normally, **~1 s** while in alarm. The gateway's **10 min / 12 h** cadence is a **separate** clock. This node is **never polled**: it broadcasts blind, the gateway's low-power scan (**100 ms every 5 s, 2% duty**) caches the latest advert, and on the gateway's own schedule it forwards that cache upward.

2. Algorithm

1. Init: load `node_id`, `wgn_group` and thresholds from NVS; start the coulomb gauge (which auto-zeroes if it detects a replacement cell).
2. Wake on the node's own RTC timer; read read aggregate (`sw_1`), `sts4x` internal temp, `bma400` peak `|a|`.
3. Evaluate against the alarm threshold **with hysteresis** (open=1 / closed=0 (debounced); impact by `|a|` threshold).
4. **Seal** the reading with AES-CCM under the per-wagon key, using a fresh monotonic counter `ctr` as the nonce (persisted in NVS so it never repeats). Nonce byte 7 carries a direction tag so an uplink nonce can never collide with a downlink one.
5. Advertise the encrypted burst (non-connectable); **slow when normal (~30 s), fast while in alarm (~1 s)**. Both periods are read at runtime from `sn_normal_s()` / `sn_fast_s()`, so a `SW_DN_SET_CADENCE` downlink changes them live and the change survives a reboot (section 5.1). A `DEBUG_TRACE` build seeds **10 s** instead of 30 s so a node is visible on the bench without waiting half a minute per

advert - the seed only, not a different mechanism.

6. Every CFGWIN_PERIOD_MS of wall-clock time - **including during an alarm** - open a 4 s connectable **config window** so the gateway can write new thresholds, or stream a firmware image (section 6).

3. Advertisement payload (encrypted `sw_adv_enc`, 21 B)

Cleartext header (company, ver, wgn_group, type, id, ctr) + **AES-CCM ciphertext** of the fields below + 4-byte MIC. The header is authenticated; the reading itself is encrypted.

Field	Meaning for this node
wgn_group	This wagon's isolation group, from the wagon number (gateway drops other groups; cleartext)
node_type / node_id	sensor type + unique id within the wagon (cleartext header)
ctr	monotonic nonce counter, persisted in NVS (cleartext; makes each advert unique)
flags / batt (encrypted)	ALARM/LOWBATT flags, battery % from the coulomb gauge
value	STS4x internal temperature ×10 (°C)
value2	peak acceleration ×100 (g) - tamper magnitude
flags	DOOR_OPEN IMPACT ALARM LOWBATT

3.1 Cadence over the air (`sw_dn_set_cadence`, 0x08)

RDSO §7.15 requires the vendor to set operating values in the field, so the advertising period is not fixed at build time. The command carries the **quiet** period in `value` and the **alarm** period in `hyst`, both in **seconds** - the same two wire fields every other threshold uses, so it inherits the individual / by-type / all-nodes fan-out unchanged.

Guard	Behaviour
Range	Both clamped to <code>CADENCE_MIN_S</code> 1 s ... <code>CADENCE_MAX_S</code> 3600 s. Below 1 s buys nothing over the alarm cadence and costs battery; above an hour the node cannot satisfy the gateway's 15-minute staleness check and would be flagged DOWN continuously.
Ordering	The alarm period is never allowed to exceed the quiet one. An alarm reporting <i>less</i> often than normal traffic would invert the purpose of the fast cadence.
Persistence	Written to NVS (<code>sw/norm</code> , <code>sw/fast</code>) and reloaded by <code>settings_load()</code> , so a cadence tuned over the air survives both a reboot and a firmware update carrying different compile-time defaults.
Effect	Immediate - the very next sleep uses the new value. No reboot.

Battery consequence: the §5.1.20 six-year projection is built on the **30 s** production figure. Dropping the quiet cadence to 5 s multiplies advertising charge by six and invalidates the life estimate - the clamp bounds the damage, it does not prevent it.

4. Alarm / event mapping

Condition	Protocol code	Notes
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Open in unauthorised zone / moving	DOOR_UNAUTH	gateway decides via geofence + motion
Authorised in-zone change	DOOR_OPEN / DOOR_CLOSE (event)	normal
Tamper impact	DOOR_TAMPER	off-spec code - not in Protocol Rev.1 §3.2
Cell low	NODE_LOW_BATTERY	advisory

5. Battery gauge (coulomb counting)

The cell is a **non-rechargeable 13 Ah Li-SOCl₂; ER34615**. Its discharge curve is flat, so a voltage-to-percent map reads ~100 % for years and then collapses. State of charge is therefore obtained by **integrating what the firmware spends**: every sleep, measurement, advertising burst and config window adds (current × time) in nAh to an accumulator persisted in NVS. Self-discharge is carried as a constant current so elapsed time is never free.

Item	Value
Usable capacity	BATT_USABLE_MAH = 10000 (derated from the 13 Ah nameplate for cold and load)
Total current ceiling	13 Ah over a 6-year life = 247 µA average
Predicted average	~134 µA (sleep 5 + self-discharge 15 + measure 1 + advertising 60 + config window 53)
Independent end-of-life check	BATT_EOL_MV = 3000 mV, so counter drift cannot hide a dying cell

Calibration outstanding: every BATT_I_*_UA constant except self-discharge is an estimate marked MEASURE: in the source. They must be measured with a Nordic PPK2 in series with the cell over one full wake cycle before the projected life means anything.

6. Firmware update over BLE

The node can be updated by **its own gateway only**. There is no SMP service, no MCUmgr, and no phone path - the image arrives as sealed chunks on a second GATT characteristic during the same config window used for thresholds.

Step	What happens
Stage	Gateway downloads the image over HTTP into its NOR ota partition and computes a CRC16. The two links are never up together, so the image must be staged before any of it can be sent.
Arm	SW_DN_OTA_BEGIN on the config characteristic: erases this node's MCUboot secondary slot and announces the size. Sent once per campaign - a resumed transfer must not repeat it.
Stream	192-byte chunks, each its own AES-CCM frame with nonce direction tag 0x02, accepted only strictly in order . Delivery spans as many windows as it takes; progress is persisted in gateway FRAM so a reset resumes rather than restarts.
Verify	SW_DN_OTA_END carries the CRC16. A mismatch discards the whole transfer before anything is marked bootable.
Swap	boot_request_upgrade(B00T_UPGRADE_TEST). No reboot here - the node swaps on its own next reset, so an update never interrupts a report in flight. MCUboot verifies the fleet signature and auto-reverts if the new image fails to confirm.

Confirm	<code>boot_write_img_confirmed()</code> , called only after the new image completes a full successful cycle - sense, seal, advertise, account. TEST mode without this step reverts on the second reboot , so an update that appeared to install would silently vanish days later. Deferring the confirm until a whole cycle has passed is what makes the auto-revert meaningful: an image that boots but cannot read its sensor is rejected by the node itself.
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Three independent gates, in order: per-chunk AES-CCM (a neighbouring wagon or an attacker cannot inject one byte), CRC16 over the whole image (catches truncation or reordering before anything is bootable), and MCUboot signature verification with auto-revert (the final authority). Gate 3 alone would stop unsigned code running, but not stop a wagon spending hours of radio assembling rubbish - that is what 1 and 2 buy.

During a campaign the gateway sends `SW_DN_FASTWIN` so windows open every few seconds instead of every 10 minutes. Fast mode is **deadlined** in the node, so an abandoned campaign or a lost cancel cannot strand it at high duty.

6. Hardware & pin map (verified against the DOOR schematic)

Function	Net	Pin
I2C SDA / SCL (BMA400 + STS4x)	SDA / SCL	P1.04 / P1.05 (i2c21)
Sensor-rail enable (TPS22917)	SW_AXI	P1.06
BMA400 interrupt INT1	BMA_INT	P0.04
Door reed aggregate RS1..RS4	SW_1	P1.11 (pull-up)
External watchdog WAKE / DONE (TPL5010)	uC_Wakeup / uC_Alive	P0.00 / P0.01
Debug UART	D_TX / D_RX	P1.02 / P1.03 (NFC pins released via <code>&uicr</code>)

The four reeds are wired in **parallel on one net**, so the firmware sees only the aggregate: the line reads "open" only when **all** reeds are open. Per-door sensing needs them split onto separate GPIOs.

7. Always-on tamper (RDSO §7.3)

§7.3 requires sensors to be "always ON to ensure capturing of any and all events". Polling once per 30 s cycle cannot do that - a tamper between samples is simply lost. On this node the sensor rail therefore **stays up** and the BMA400's own motion interrupt wakes the MCU, so an impact is timed by hardware rather than by the sample clock.

Applied to DOOR_B1D_L only. The interrupt-driven path (`impactint.c`) has not yet been propagated to the other three DOOR nodes, which still use a blind sleep between samples.

8. Open items

- **STS4x address:** the firmware uses `0x44`; the generated schematic pin-map sheet says `0x4A`. Confirm the exact STS4x variant - one of the two is wrong.
- **BMA400 INT1** configuration was corrected against the datasheet (activity + OR, non-latched, push-pull, active high) but has never been confirmed on hardware: verify it fires on a knock and *not* on stillness.
- `DOOR_TAMPER` is not a Protocol Rev.1 alarm code.